Module 09 • API Reference

Complete REST API Reference

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

Complete REST API Reference

The IRRIGATEWISE application serves both legacy endpoints (for backward compatibility) and the upgraded `/api/v1/*` enterprise IoT & Machine Learning API.

Base URL: `http://localhost:8000` **Interactive Swagger UI:** `http://localhost:8000/docs` **ReDoc Specification:** `http://localhost:8000/redoc`

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2. IoT Sensor Ingestion & Telemetry (`/api/v1/sensors`)

`POST /api/v1/sensors/readings`

Ingests an edge telemetry packet from ESP32 or simulator. Automatically screens for sensor anomalies before committing to database. - **Request Body:** ```json

"node_id": "ESP32-NODE-01", "plot_id": "hydrogel_plot", "soil_moisture": 49.2, "temperature": 32.1, "humidity": 64.0, "rainfall": 0.0, "hydrogel_presence": 1, "hydrogel_dosage": 50.0, "pump_status": "OFF"

`` - **Response`200 OK`:** Ingestion receipt and anomaly screening status.

`GET /api/v1/sensors/latest`

Returns the most recent verified sensor telemetry packet.

`GET /api/v1/sensors/history?limit=24`

Returns time-series sensor history for chart plotting.

`GET /api/v1/sensors/anomalies`

Returns flagged sensor anomaly events and failure classifications.

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3. Machine Learning Intelligence (`/api/v1/ml`)

`POST /api/v1/ml/moisture/predict`

Executes Multi-Horizon Gradient Boosting inference for 1h, 3h, and 6h future soil moisture. - **Request Body:** ```json

"soil_moisture": 48.0, "temperature": 32.0, "humidity": 65.0, "rainfall": 0.0, "hour_of_day": 14, "hydrogel_presence": 1, "hydrogel_dosage": 50.0

`- **Response `200 OK`:**` json

"current_moisture": 48.0, "predicted_1h": 46.85, "predicted_3h": 44.12, "predicted_6h": 42.90, "target_moisture": 62.0, "confidence_score": 0.88, "model_name": "Gradient Boosting Multi-Output Regressor", "model_version": "v1.2.0-gb-prod"

```

`POST /api/v1/ml/crop/predict`

Predicts crop suitability using the Random Forest classifier (87.5% accuracy). - **Request Body:** ```json

"nitrogen": 85.0, "phosphorus": 42.0, "potassium": 210.0, "ph": 6.8, "organic_carbon": 0.82, "temperature": 31.0, "humidity": 70.0, "rainfall": 100.0, "soil_type": "loam"

`- **Response `200 OK`:**` json

"recommended_crop": "Cotton", "confidence": 0.82, "top_candidates": [ "crop": "Cotton", "probability": 0.82 , "crop": "Maize", "probability": 0.11 , "crop": "Groundnut", "probability": 0.07 ], "hydrogel_compatibility": "High (Recommended dosage: 50 g/m$^2$)", "notes": "Excellent nutrient balance for deep-taproot fibrous growth."

```

`GET /api/v1/ml/models`

Returns active production model metadata and all registered versions with verified MAE/RMSE/R^2 scores.

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4. Predictive Irrigation & Actuator Control (`/api/v1/irrigation`)

`POST /api/v1/irrigation/predict`

Evaluates field telemetry across both **Mode 1 (Rule-Based SMD)** and **Mode 2 (Autonomous ML Predictive)**. - **Response** `200 OK` **:** Side-by-side comparison including water savings percentage and rain suppression flags.

`POST /api/v1/irrigation/execute`

Dispatches an irrigation pulse to the physical relay or hardware simulator, subject to safety interlocks. - **Request Body:** ```json

"mode": "ml_predictive", "duration_minutes": 10.0, "water_volume_liters": 1500.0, "override_safety": false

`` - **Response` 200 OK`:** Pump status transition and execution receipt.

`GET /api/v1/irrigation/pump-status`

Returns real-time hardware status (`OFF` , `ON` , `COOLDOWN` , `LOCKED`).

`GET /api/v1/irrigation/feedback`

Returns closed-loop verification metrics comparing historical predictions against observed ground truth.

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5. Reports & Data Export (`/api/v1/reports`)

`GET /api/v1/reports/summary`

Returns live aggregated metrics: total soil samples, mean fertility, hydrogel water saved, and total irrigation events.

`GET /api/v1/reports/export/soil-csv`

Downloads complete soil testing history in RFC 4180 CSV format.

`GET /api/v1/reports/export/irrigation-csv`

Downloads complete irrigation audit history in RFC 4180 CSV format.